Contents

1 Order and Degree of a Differential Equation

Once you have identified an equation as a Differential Equation (DE), the next critical step is to classify it by its **Order** and **Degree**. These two properties determine the complexity of the equation and dictate which mathematical methods you will need to use to solve it.

1.1 1. Concept Overview

1.1.1 1.1 The Order of a Differential Equation

> **Definition:** The **Order** of a differential equation is the order of the highest differential coefficient (derivative) present in the equation.

- If the highest derivative is $\frac{dy}{dx}$ (or y'), it is a **1st-order** DE.
- If the highest derivative is $\frac{d^2y}{dx^2}$ (or y''), it is a **2nd-order** DE.
- If the highest derivative is $\frac{d^ny}{dx^n}$ (or $y^{(n)}$), it is an **n-th order** DE.

1.1.2 1.2 The Degree of a Differential Equation

> **Definition:** The **Degree** of a differential equation is the highest power (exponent) of the highest-order derivative present in the equation.

1.2 2. Important Working Rule

To accurately find the degree of a differential equation, the equation **must be made free from radicals and fractions** as far as the derivatives are concerned.

If a derivative is inside a square root or raised to a fractional power (e.g., 3/2), you must use algebraic manipulation (like squaring or cubing both sides) to clear those fractional powers before determining the degree.

1.3 3. Solved Questions

Here are step-by-step examples from the lecture notes demonstrating how to find the order and degree.

1.3.1 Question 1

Determine the order and degree of the following Differential Equation:

$$\cos x\frac{d^2y}{dx^2} + \sin x\left(\frac{dy}{dx}\right)^3 + 8y = \tan x$$

Solution:

1. **Find the Order:** Look for the highest derivative in the equation.
   - We have a first derivative: $\frac{dy}{dx}$
   - We have a second derivative: $\frac{d^2y}{dx^2}$
   - The highest derivative is the second derivative. Therefore, the **Order is 2**.

2. **Find the Degree:** Look at the power of the *highest* derivative.
   - The highest derivative is $\frac{d^2y}{dx^2}$.
   - It is not raised to any explicitly written power, which means its power is 1.
   - *(Note: Do not be tricked by the* $\left(\frac{dy}{dx}\right)^3$ *term. Even though it has a higher power of 3, it is not the highest-order derivative).*
3. **Final Answer:**
   - **Order:** 2
   - **Degree:** 1

1.3.2 Question 2

Determine the order and degree of the following Differential Equation:

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = K\frac{d^2y}{dx^2}$$

Solution:

1. **Apply the Working Rule (Clear Radicals):** We notice that the derivatives on the left side are raised to a fractional power (3/2). We cannot determine the degree until this fraction is removed.
2. **Algebraic Manipulation:** To eliminate the denominator of the fraction (/2), we square both sides of the equation.

$$\left(\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}\right)^2 = \left(K\frac{d^2y}{dx^2}\right)^2$$

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = K^2\left(\frac{d^2y}{dx^2}\right)^2$$

3. **Find the Order:** Now look at the manipulated equation. The highest derivative present is $\frac{d^2y}{dx^2}$.
   - Therefore, the **Order is 2**.
4. **Find the Degree:** Look at the power of that highest derivative.
   - The term $\left(\frac{d^2y}{dx^2}\right)$ is raised to the power of 2.
   - Therefore, the **Degree is 2**.
5. **Final Answer:**
   - **Order:** 2
   - **Degree:** 2

Next Topic: Now that you know how to classify DEs, proceed to Topic 1.3: Formation of Differential Equations to learn how these equations are originally constructed.